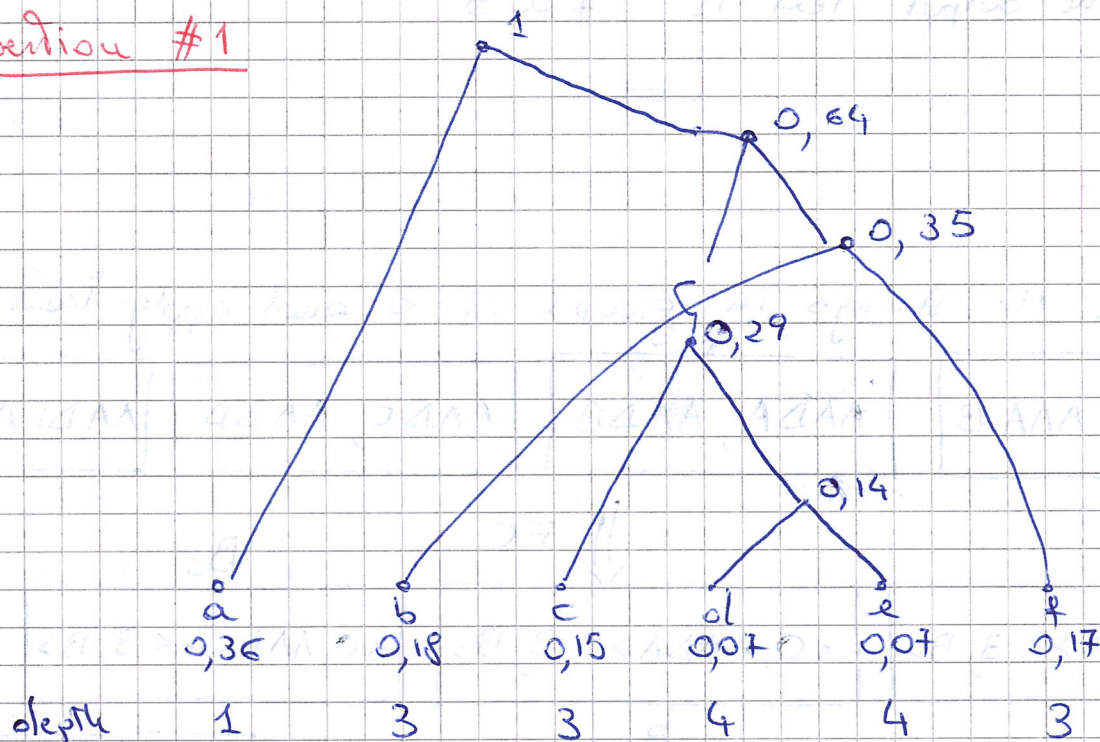


Algorithm Engineering 10/1/25

Question #1



Num	1	2	3	4
	1	0	3	2

Symb	
1	a
2	//
3	b, c, f
4	d, e

FC	1	2	3	4
	1	2	1	0

the compressed input is: 0 1 1 1 0 0 1
f a

$l = 1$; $v = 0$

$v \nless FC[1] \rightarrow$ ~~output [1,0] = a~~

$l = 2$; $v = 1$; $v < FC[2]$

$l = 3$; $v = 3$; $v \geq FC[3] \rightarrow$ output Symb[3,2] = f

$l = 1$; $v = 1$; $v \geq FC[1] \rightarrow$ output Symb[1,0] = a

$l = 1$; $v = 0$; $v < FC[1]$

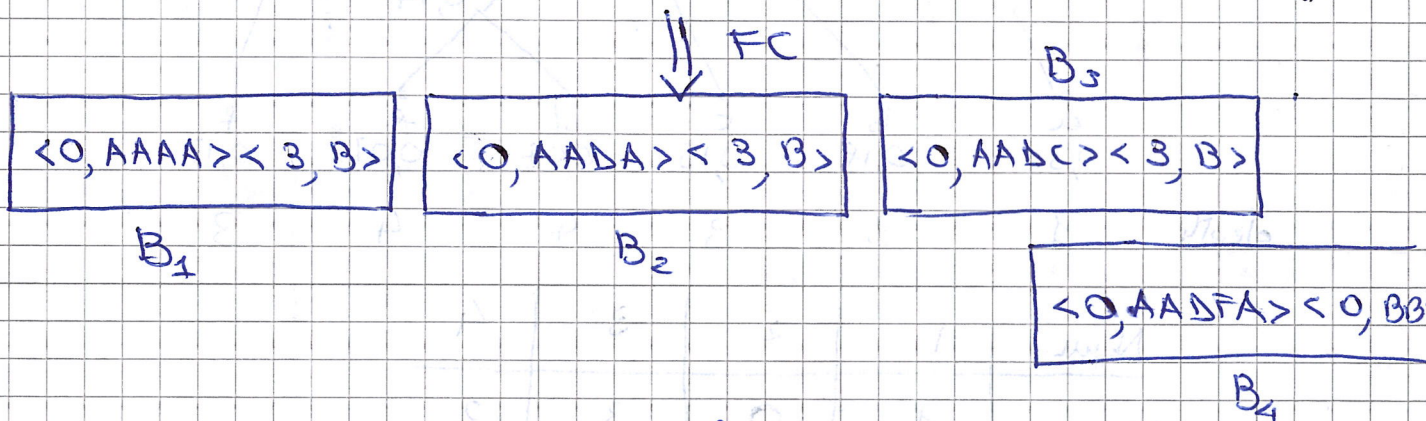
$l = 2$; $v = 0$; $v < FC[2]$;

$x = 3; v = 1; v \geq FC[D] \rightarrow \text{output } \text{SymB}[3, 0] = b$

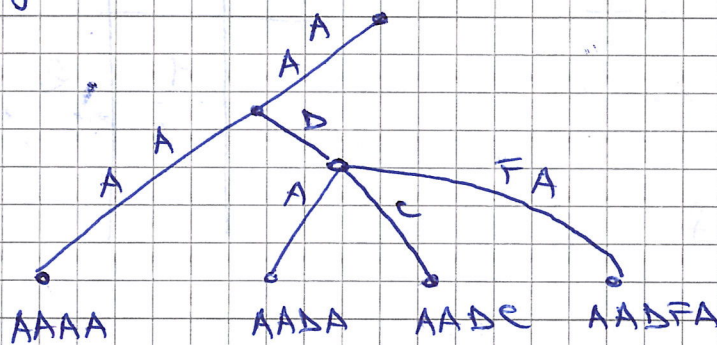
therefore the output text is: $\# a b$

Question #2

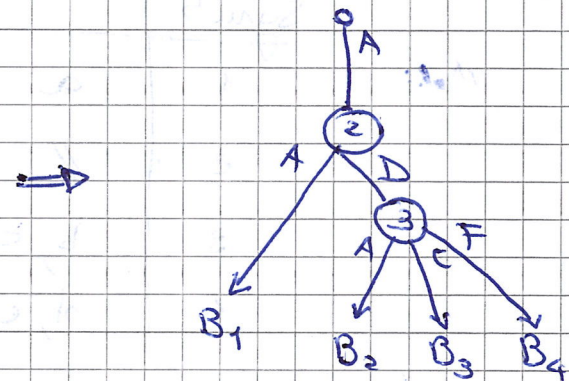
We block the strings in groups of 2 and apply Fred Coding



We index in memory via a Patricia tree the first string of every block B_i .



Compressed tree



Patricia tree

Prefix Searching for $P = AAD$, counts of searching for $AAD \$$ and $AAD \#$.

Both searches end up in node $\textcircled{3}$ and thus they may have the same string, say in $B_2 \rightarrow AADA$ and compute the same $lcp = 3$, and mismatch 'A'.

But AAD\$ ends up leftmost, while AAD# ends up rightmost among the children of



because

\$ is the smallest symbol and # is the largest symbol.

So AAD\$ scans block B_1 , up to its end.

AA# " " B_4 , and stops before B_3 .

So the search returns the entire blocks B_2, B_3 and the first string of B_4 .

Question 3

$$s=1, c=3 \rightarrow s=\{00\}, c=\{01, 10, 11\}$$

the first 7 codewords are

s	00			ccs	01	01	00
cs	01	00			01	10	00
	10	00			01	11	00
	11	00					

In the case of interpolative code for $(1, 4, 5, 6, 9, 11, 15)$

$$l=1, low=1, r=7, hi=15$$

•) $m=4, sm=6$

$$range = [4, 12] \quad \#bits = \lceil \lg_2(12-4+1) \rceil = 4$$

$$encode\ 2\ in\ 4\ bits \Rightarrow 0010$$

•) $l=1, low=1, r=3, hi=5 \rightarrow sm=4$

$$range = [2, 4] \quad \#bits = \lceil \lg_2(4-2+1) \rceil = 2$$

$$encode\ 2\ in\ 2\ bits \Rightarrow 10$$

$$l = 4, \text{ low} = 7, \text{ h} = 7, \text{ hi} = 15 \rightarrow \text{sum} = 11$$

$$\text{range} = [8, 14] \quad \# \text{ bits} = \left\lceil \log_2 (14 - 8 + 1) \right\rceil = 3$$

encode 3 in 3 bits $\Rightarrow 011$

Question #4

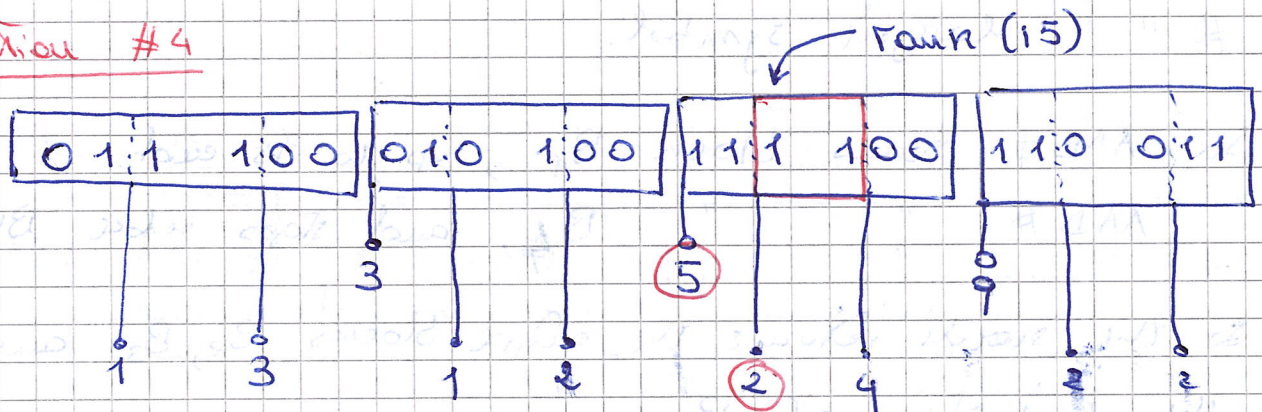


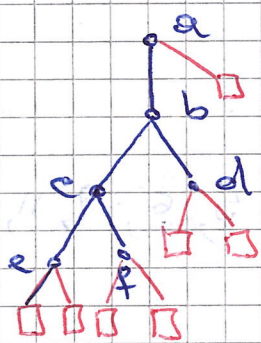
Table =

	0	1
00	0	0
01	0	1
10	1	1
11	1	2

rank (15) = rank-bj-block (3)
 + rank-local₃ (2)
 + T[11, 1]
 = 5 + 2 + 1 = 8

in red

Question #5



1 2 3 4 5 6
 $B = 110111000000$
 $L = [a, b, c, d, e, f]$
 1 2 3 4 5 6

checking for the path [a, f] starting from the root:

$L[1] = 'a' \rightarrow \text{ok}$

$\text{right}(1) = r(2 \cdot 1 + 1) \Rightarrow$ the right child does not exist
 $B[3] = 0$

$\text{left}(1) = \text{rank}(2 \cdot 1) = 2 \rightarrow$ the left child does exist
 $B[2] = 1$

and its label is $L[2] = b \neq f$

So the path does not exist.